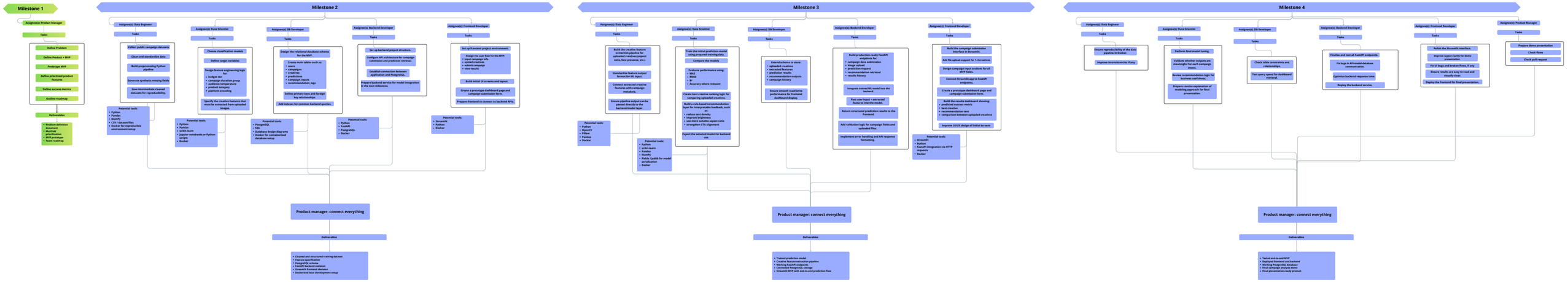
Milestone 1
Assignee(s): Product Manager
Tasks
Define Problem
Define Product + MVP
Prototype MVP
Define prioritised product features
Define success metrics
Outline roadmap
Deliverables
• Problem definition document
• Prioritised
• prioritisation
• MVP prototype
• Team roadmap
Milestone 2
Assignee(s): Data Engineer
Tasks
Collect public campaign datasets
Clean and standardise data
Build preprocessing Python pipeline
Generate synthetic missing fields
Save intermediate cleaned datasets for reproducibility.
Potential tools:
• Python
• Pandas
• NumPy
• Parquet files
• Docker for reproducible environment setup
Assignee(s): Data Scientist
Tasks
Choose classification models
Define target variables
Design feature engineering logic for:
• budget tier
• campaign duration group
• audience temperature
• creative type
• platform encoding
Specify the creative features that must be extracted from uploaded images.
Potential tools:
• Python
• Pandas
• Scikit-learn
• Jupyter notebooks or Python scripts
• Docker
Assignee(s): DB Developer
Tasks
Design the relational database schema for the MVP.
Create main tables such as:
• users
• campaigns
• creatives
• predictions
• campaign_inputs
• recommendation_logs
Define primary keys and foreign key relationships.
Add indexes for common backend queries.
Potential tools:
• PostgreSQL
• SQL
• Excalidraw design diagrams
• Docker for containerised database setup
Assignee(s): Backend Developer
Tasks
Set up backend project structure.
Configure API architecture for campaign submission and prediction retrieval.
Establish connection between application and PostgreSQL.
Prepare backend service for model integration in the next milestone.
Potential tools:
• Python
• FastAPI
• PostgreSQL
• Docker
Assignee(s): Frontend Developer
Tasks
Set up frontend project environment.
Design the user flow for the MVP:
• input campaign info
• upload creatives
• submit campaign
• view results
Build initial UI screens and layout.
Create a prototype dashboard page and campaign submission form.
Prepare frontend to connect to backend APIs.
Potential tools:
• Streamlit
• Python
• Docker
Product manager: connect everything
Deliverables
• Cleaned and structured training dataset
• Feature engineering
• PostgreSQL schema
• FastAPI backend skeleton
• Streamlit frontend skeleton
• Dockerised local development setup
Milestone 3
Assignee(s): Data Engineer
Tasks
Build the creative feature extraction pipeline: uploaded creatives (paper) ratio, face presence, etc.)
Standardise feature output format for ML input.
Connect extracted creative features with campaign metadata.
Ensure pipeline output can be reused by both the backend/model layer.
Potential tools:
• Python
• OpenCV
• Pillow
• Pandas
• Docker
Assignee(s): Data Scientist
Tasks
Train the initial prediction model using prepared training data.
Compare the models
Evaluate performance using:
• MAE
• RMSE
• R²
• Accuracy where relevant
Create best creative ranking logic for comparing uploaded creatives.
Build a rule-based recommendation layer for interpretable feedback, such as:
• reduce text density
• improve brightness
• use more suitable aspect ratio
• strengthen CTA alignment
Export the selected model for backend use.
Potential tools:
• Python
• Scikit-learn
• Pandas
• NumPy
• Pickle / joblib for model serialization
• Docker
Assignee(s): DB Developer
Tasks
Extend schema to store:
• uploaded creatives
• extracted features
• prediction results
• recommendation outputs
• campaign history
Ensure smooth read/write performance for frontend dashboard display.
Assignee(s): Backend Developer
Tasks
Build production-ready FastAPI endpoints for:
• campaign submission
• image upload
• prediction request
• recommendation retrieval
• results history
Integrate trained ML model into the backend.
Pass user input + extracted features into the model.
Return structured prediction results to the frontend.
Add validation logic for campaign fields and uploaded files.
Implement error handling and API response formatting.
Assignee(s): Frontend Developer
Tasks
Build the campaign submission interface in Streamlit.
Add file upload support for 1-3 creatives.
Design campaign input sections for all MVP fields.
Connect Streamlit app to FastAPI endpoints.
Create a prototype dashboard page and campaign submission form.
Build the results dashboard showing:
• predicted performance metric
• best creative
• recommendation text
• comparison between uploaded creatives
Improve UI/UX design of initial screens.
Potential tools:
• Streamlit
• Python
• FastAPI integration via HTTP requests
• Docker
Product manager: connect everything
Deliverables
• Trained prediction model
• Creative feature extraction pipeline
• Working FastAPI endpoints
• Working PostgreSQL database
• Streamlit MVP with end-to-end prediction flow
Milestone 4
Assignee(s): Data Engineer
Tasks
Ensure reproducibility of the data pipeline in Docker.
Improve inconsistencies if any
Assignee(s): Data Scientist
Tasks
Perform final model tuning.
Validate whether outputs are meaningful for each campaign intent.
Review recommendation logic for business usefulness.
Prepare concise explanation of modeling approach for final presentation.
Assignee(s): DB Developer
Tasks
Check table constraints and relationships.
Test query speed for dashboard/ overview.
Assignee(s): Backend Developer
Tasks
Finalize and test all FastAPI endpoints.
Fix bugs in API-model-database communication.
Optimize backend response time.
Deploy the backend service.
Assignee(s): Frontend Developer
Tasks
Polish the Streamlit interface.
Improve layout clarity for demo presentation.
Fix UI bugs and broken flows, if any.
Ensure results are easy to read and visually clear.
Deploy the frontend for final presentation.
Assignee(s): Product Manager
Tasks
Prepare demo presentation
Check flows
Check pull request
Product manager: connect everything
Deliverables
• Tested end-to-end MVP
• Deployed frontend and backend
• Working PostgreSQL database
• Stable demo workflow
• Final presentation-ready product